Question 1 (5 marks)	Evaluate the following (i) $\left(\frac{625}{81}\right)^{-\frac{3}{4}}$ ii) Simplify the following as much as possible $$\frac{\mathbf{30}\left(\boldsymbol{p}^{-\mathbf{6}}\right)^{-\mathbf{3}}\left(\boldsymbol{q}^{\mathbf{5}}\right)^{-\mathbf{4}}\boldsymbol{r}^{\mathbf{3}}}{\mathbf{15}\boldsymbol{p}^{-\mathbf{5}}\boldsymbol{q}^{\mathbf{4}}\sqrt{\boldsymbol{r}}}$$
Question 2 (5 marks)	Solve the following for x: (i) $\frac{2}{7^x} = 98.\sqrt{7}$ (ii) $x^{\frac{-3}{2}} = 8$

Question 3 (3 marks)	$Solve \quad 2^{t^3-24} = 8$ for two values of t
Question 4 (3 marks)	Solve the following for x $\sqrt[4]{\frac{49^{x+1}}{243^x}} = 7$